Quantum Field Theory – Homework 2 (Solutions)

Name: ______________________ **Due Date:** 2025.11.03

Problem 1. Probability Interpretation and the Klein–Gordon Equation (20 points)

In non-relativistic quantum mechanics, the Schrödinger equation

$$i\frac{\partial}{\partial t}\psi(\vec{x},t) = -\frac{1}{2m}\nabla^2\psi(\vec{x},t)$$

admits a conserved quantity interpreted as probability. Multiplying by ψ^* and subtracting the complex conjugate gives

$$\frac{\partial}{\partial t}|\psi|^2 + \nabla\cdot\left[\frac{i}{2m}\big(\psi\nabla\psi^* - \psi^*\nabla\psi\big)\right] = 0.$$

Hence we can identify

$$\rho = |\psi|^2, \qquad \vec{J} = \frac{i}{2m}\big(\psi\nabla\psi^* - \psi^*\nabla\psi\big),$$

which satisfy the continuity equation $\partial_t\rho + \nabla\cdot\vec{J} = 0$, so that ρ is real, non-negative, and represents the probability density.

Now consider the relativistic Klein–Gordon equation for a single relativistic particle

$$\partial_\mu\partial^\mu\psi(\vec{x},t) - m^2\psi(\vec{x},t) = 0.$$

(a) Derive the conserved current associated with this equation by combining it with its complex conjugate. Show that it can be written as

$$\partial_\mu j^\mu = 0, \qquad j^\mu = i\big(\psi^*\partial^\mu\psi - \psi\,\partial^\mu\psi^*\big).$$

Identify the time component j^0 and spatial component $\vec{j}$.

Starting from the Klein–Gordon equation

$$(\partial_\mu\partial^\mu - m^2)\psi = 0,$$

and its complex conjugate

$$(\partial_\mu\partial^\mu - m^2)\psi^* = 0,$$

multiply the first by ψ^* and the second by ψ and subtract:

$$\psi^*(\partial_\mu\partial^\mu\psi) - \psi(\partial_\mu\partial^\mu\psi^*) = 0.$$

Rewriting gives

$$\partial_\mu\big(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*\big) = 0.$$

Hence the conserved current is

$$\boxed{j^\mu = i\big(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*\big) \qquad \text{with} \quad \partial_\mu j^\mu = 0.}$$

The time component and spatial part are

$$j^0 = i(\psi^*\dot{\psi} - \psi\dot{\psi}^*), \qquad \vec{j} = i(\psi^*\nabla\psi - \psi\nabla\psi^*).$$

(b) Examine the expression for j^0 and discuss its sign. Why can j^0 take negative values even for well-behaved wave packets? Explain why it therefore cannot be interpreted as a probability density.

The quantity $j^0 = i(\psi^*\dot{\psi} - \psi\dot{\psi}^*)$ is not positive-definite. For plane-wave solutions $\psi \sim e^{-iEt+i\vec{k}\cdot\vec{x}}$, we have

$$j^0 = 2E|\psi|^2.$$

For positive-frequency modes $E > 0$, $j^0 > 0$, but for negative-frequency solutions $E < 0$, $j^0 < 0$. Hence, j^0 can take either sign and cannot be interpreted as a probability density.

(c) Briefly explain the physical implication of this result. Why does the failure of a positive-definite probability density indicate that the Klein-Gordon equation cannot consistently describe a single relativistic particle, and motivates the need for a field-theoretic description?

Because j^0 is not positive definite, the Klein–Gordon equation cannot describe the probability density of a single relativistic particle. Instead, ψ must be interpreted as a **field**, not a wavefunction, whose excitations correspond to particles and antiparticles. This failure motivates the transition from relativistic quantum mechanics to **quantum field theory**.

Problem 2. Noethers Theorem and Energy-Momentum Conservation (20 points)

Consider the real scalar field theory

$$S[\phi] = \int d^4x \left[-\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi)\right].$$

(a) Show that the action is invariant under spacetime translations $x^\mu \to x^\mu + a^\mu$.

We consider the real scalar field with Lagrangian density

$$\mathcal{L}(\phi, \partial_\mu\phi) = -\frac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - V(\phi).$$

A spacetime translation is defined by

$$x^\mu \to x'^\mu = x^\mu + a^\mu, \qquad a^\mu = \text{constant}.$$

A scalar field satisfies

$$\phi'(x') = \phi(x).$$

Expanding to first order in the infinitesimal parameter a^μ, we obtain

$$\phi'(x) = \phi(x-a) = \phi(x) - a^\nu\partial_\nu\phi(x),$$

so the field variation at fixed x is

$$\delta\phi(x) = \phi'(x) - \phi(x) = -a^\nu\partial_\nu\phi(x).$$

The measure is invariant under translations:

$$d^4x' = d^4x.$$

Since the Lagrangian contains no explicit dependence on x^μ,

$$\mathcal{L}'(\phi', \partial\phi') = \mathcal{L}(\phi, \partial\phi).$$

Thus the action transforms as

$$S' = \int d^4x\,\mathcal{L}' = \int d^4x\,\mathcal{L} = S,$$

which shows that the action is invariant under spacetime translations.

(b) Using Noethers theorem, derive the conserved current associated with this symmetry and identify the corresponding conserved quantity.

Under a general spacetime symmetry, both the field and coordinates vary:

$$\delta x^\mu = a^\mu, \qquad \delta\phi = -a^\nu \partial_\nu \phi.$$

Derivation of the Noether current for spacetime symmetries. For spacetime transformations, the variation of the action contains two pieces:

$$\delta S = \int d^4x \, (\delta\mathcal{L} + \mathcal{L}\,\partial_\mu \delta x^\mu)\,.$$

The variation of the Lagrangian at fixed x is

$$\hat{\delta}\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi}\,\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu(\delta\phi).$$

Integrating the second term by parts and using the Euler–Lagrange equations gives

$$\delta S = \int d^4x \, \partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi + \mathcal{L}\,\delta x^\mu\right].$$

If the transformation is a symmetry, then $\delta S = 0$ for arbitrary integration regions, so the bracket must be a conserved current:

$$\boxed{J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi + \mathcal{L}\,\delta x^\mu.}$$

Applying this to translations. For a translation in the ν-direction:

$$a^\mu = \epsilon\,\delta^\mu{}_\nu, \qquad \delta x^\mu = \epsilon\,\delta^\mu{}_\nu, \qquad \delta\phi = -\epsilon\,\partial_\nu\phi.$$

Substituting:

$$J^\mu_{(\nu)} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}(-\partial_\nu\phi) + \mathcal{L}\,\delta^\mu{}_\nu.$$

For our scalar Lagrangian,

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = -\partial^\mu\phi,$$

so

$$J^\mu_{(\nu)} = \partial^\mu\phi\,\partial_\nu\phi - \delta^\mu{}_\nu\,\mathcal{L}.$$

Thus the Noether current is

$$\boxed{T^\mu{}_\nu = \partial^\mu\phi\,\partial_\nu\phi - \delta^\mu{}_\nu\,\mathcal{L}, \qquad \partial_\mu T^\mu{}_\nu = 0.}$$

The conserved charges are

$$Q_{(\nu)} = \int d^3x\, T^0{}_\nu.$$

These correspond to:

$$Q_{(0)} = \text{total energy}, \qquad Q_{(i)} = \text{components of total momentum}.$$

(c) Write down the components of the energy-momentum tensor $T^{\mu\nu}$ for this theory.

From the expression above,

$$T^\mu{}_\nu = \partial^\mu\phi\,\partial_\nu\phi - \delta^\mu{}_\nu\,\mathcal{L}.$$

Raising the second index:

$$\boxed{T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - g^{\mu\nu}\left[\tfrac{1}{2}(\partial_\alpha\phi)(\partial^\alpha\phi) + V(\phi)\right].}$$

Explicitly:

$$\begin{aligned}
T^{00} &= \tfrac{1}{2}(\partial_0\phi)^2 + \tfrac{1}{2}(\nabla\phi)^2 + V(\phi),\\
T^{0i} &= \partial^0\phi\,\partial^i\phi,\\
T^{ij} &= \partial^i\phi\,\partial^j\phi - \delta^{ij}\left[\tfrac{1}{2}(\partial_\alpha\phi)(\partial^\alpha\phi) + V(\phi)\right].
\end{aligned}$$

Problem 3. Lorentz Invariance of the Scalar Field Action (15 points)

Show explicitly that the scalar field action

$$S = \int d^4x \left[-\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2\right]$$

is invariant under an infinitesimal Lorentz transformation

$$x^\mu \to x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \qquad \phi(x) \to \phi(x) = \phi(x),$$

where $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$.

We want to show that

$$S[\phi] = \int d^4x \left[-\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2\right]$$

is invariant under the infinitesimal Lorentz transformation

$$x^\mu \to x'^\mu = x^\mu + \Lambda^\mu{}_\nu x^\nu, \qquad \phi(x) \to \phi'(x') = \phi(x),$$

with $\Lambda_{\mu\nu} = -\Lambda_{\nu\mu}$ and $|\Lambda^\mu{}_\nu| \ll 1$.

1. Transformation of the measure. We first compute the Jacobian of the coordinate change:

$$\frac{\partial x'^\mu}{\partial x^\nu} = \delta^\mu{}_\nu + \Lambda^\mu{}_\nu.$$

Then

$$\det\left(\frac{\partial x'}{\partial x}\right) = \det(\delta^\mu{}_\nu + \Lambda^\mu{}_\nu) = 1 + \Lambda^\mu{}_\mu + \mathcal{O}(\Lambda^2).$$

Because $\Lambda_{\mu\nu}$ is antisymmetric,

$$\Lambda^\mu{}_\mu = \eta^{\mu\rho}\Lambda_{\rho\mu} = -\eta^{\mu\rho}\Lambda_{\mu\rho} = 0.$$

Hence

$$\det\left(\frac{\partial x'}{\partial x}\right) = 1 + \mathcal{O}(\Lambda^2),$$

so, to first order in Λ,

$$d^4x' = d^4x.$$

Thus the measure is invariant at this order.

2. Transformation of the field and its derivatives. The field is a scalar, so

$$\phi'(x') = \phi(x).$$

For later convenience, it is useful to have the inverse transformation:

$$x^\mu = x'^\mu - \Lambda^\mu{}_\nu x'^\nu + \mathcal{O}(\Lambda^2).$$

Then

$$\phi'(x') = \phi(x) = \phi\big(x' - \Lambda x'\big) = \phi(x') - \Lambda^\rho{}_\sigma x'^\sigma \partial_\rho \phi(x') + \mathcal{O}(\Lambda^2).$$

Therefore, at fixed x',

$$\delta\phi(x') \equiv \phi'(x') - \phi(x') = -\Lambda^\rho{}_\sigma x'^\sigma \partial_\rho \phi(x').$$

Next, the derivatives transform via the chain rule. The derivative with respect to x' is

$$\partial'_\mu \equiv \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial}{\partial x^\rho}.$$

Using

$$x^\rho = x'^\rho - \Lambda^\rho{}_\sigma x'^\sigma,$$

we obtain

$$\frac{\partial x^\rho}{\partial x'^\mu} = \delta^\rho{}_\mu - \Lambda^\rho{}_\mu,$$

so to first order

$$\partial'_\mu = \left(\delta^\rho{}_\mu - \Lambda^\rho{}_\mu\right) \partial_\rho.$$

Now act on the scalar field:

$$\partial'_\mu \phi'(x') = \left(\delta^\rho{}_\mu - \Lambda^\rho{}_\mu\right) \partial_\rho \phi(x).$$

To first order we can identify $x \approx x'$, so

$$\partial'_\mu \phi'(x') = \left(\delta^\rho{}_\mu - \Lambda^\rho{}_\mu\right) \partial_\rho \phi(x') + \mathcal{O}(\Lambda^2).$$

3. Invariance of the kinetic term. The kinetic term in the transformed action is

$$\partial'_\mu \phi'(x')\, \partial'^\mu \phi'(x') = \eta^{\mu\nu}\, \partial'_\mu \phi'(x')\, \partial'_\nu \phi'(x').$$

Substitute the expression for $\partial'_\mu\phi'$:

$$\begin{aligned}\partial'_\mu\phi'\,\partial'_\nu\phi' &= (\delta^\rho{}_\mu - \Lambda^\rho{}_\mu)(\delta^\sigma{}_\nu - \Lambda^\sigma{}_\nu)\partial_\rho\phi\,\partial_\sigma\phi \\ &= \left[\delta^\rho{}_\mu\delta^\sigma{}_\nu - \Lambda^\rho{}_\mu\delta^\sigma{}_\nu - \delta^\rho{}_\mu\Lambda^\sigma{}_\nu + \mathcal{O}(\Lambda^2)\right]\partial_\rho\phi\,\partial_\sigma\phi.\end{aligned}$$

Thus

$$\begin{aligned}\partial'_\mu\phi'\,\partial'^\mu\phi' &= \eta^{\mu\nu}\partial'_\mu\phi'\,\partial'_\nu\phi' \\ &= \eta^{\mu\nu}\left[\delta^\rho{}_\mu\delta^\sigma{}_\nu - \Lambda^\rho{}_\mu\delta^\sigma{}_\nu - \delta^\rho{}_\mu\Lambda^\sigma{}_\nu\right]\partial_\rho\phi\,\partial_\sigma\phi + \mathcal{O}(\Lambda^2) \\ &= \eta^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi - \eta^{\mu\nu}\Lambda^\rho{}_\mu\delta^\sigma{}_\nu\partial_\rho\phi\,\partial_\sigma\phi - \eta^{\mu\nu}\delta^\rho{}_\mu\Lambda^\sigma{}_\nu\partial_\rho\phi\,\partial_\sigma\phi + \mathcal{O}(\Lambda^2).\end{aligned}$$

We now simplify the second and third terms. Using

$$\eta^{\mu\nu}\Lambda^\rho{}_\mu\delta^\sigma{}_\nu = \Lambda^{\rho\sigma}, \qquad \eta^{\mu\nu}\delta^\rho{}_\mu\Lambda^\sigma{}_\nu = \Lambda^{\sigma\rho},$$

we get

$$\partial'_\mu\phi'\,\partial'^\mu\phi' = \partial_\rho\phi\,\partial^\rho\phi - \Lambda^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi - \Lambda^{\sigma\rho}\partial_\rho\phi\,\partial_\sigma\phi + \mathcal{O}(\Lambda^2).$$

But $\Lambda^{\rho\sigma} = -\Lambda^{\sigma\rho}$ is antisymmetric, while $\partial_\rho\phi\,\partial_\sigma\phi$ is symmetric in (ρ,σ). Therefore the contractions vanish:

$$\Lambda^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi = 0, \qquad \Lambda^{\sigma\rho}\partial_\rho\phi\,\partial_\sigma\phi = 0.$$

So we obtain

$$\partial'_\mu\phi'\,\partial'^\mu\phi' = \partial_\mu\phi\,\partial^\mu\phi + \mathcal{O}(\Lambda^2).$$

Thus the kinetic term is invariant to first order in Λ.

4. Invariance of the mass term. Since $\phi'(x') = \phi(x)$, we have

$$\phi'^2(x') = \phi^2(x),$$

so the mass term $m^2\phi^2$ is clearly invariant:

$$m^2\phi'^2(x') = m^2\phi^2(x).$$

5. Putting it all together. The transformed action is

$$S' = \int d^4x'\left[-\tfrac{1}{2}\partial'_\mu\phi'\,\partial'^\mu\phi' - \tfrac{1}{2}m^2\phi'^2\right].$$

Changing variables back to x and using $d^4x' = d^4x$ (to first order), and the invariance of both the kinetic and mass terms, we get

$$S' = \int d^4x \left[-\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2\right] = S.$$

Therefore $\delta S = S' - S = 0$ to first order in Λ, and the scalar field action is invariant under infinitesimal Lorentz transformations.

Then, use Noether's theorem to find the expression for the conserved angular-momentum current $M^{\lambda\mu\nu}$.

We now derive the conserved current associated with Lorentz invariance and show that it can be written as

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

1. Infinitesimal Lorentz transformation. An infinitesimal Lorentz transformation is

$$x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu, \qquad \omega_{\mu\nu} = -\omega_{\nu\mu},$$

where $\omega_{\mu\nu}$ are small, antisymmetric parameters. For a scalar field, we have

$$\phi'(x') = \phi(x).$$

As before, expanding to first order gives the field variation at fixed x:

$$\delta\phi(x) \equiv \phi'(x) - \phi(x) = -\omega^\rho{}_\sigma x^\sigma \partial_\rho\phi(x).$$

Similarly,

$$\delta x^\mu = \omega^\mu{}_\nu x^\nu.$$

2. Noether current for spacetime symmetries. From our general result for spacetime symmetries, the Noether current is

$$J^\lambda = \frac{\partial\mathcal{L}}{\partial(\partial_\lambda\phi)}\,\delta\phi + \mathcal{L}\,\delta x^\lambda.$$

For the scalar field Lagrangian

$$\mathcal{L} = -\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2,$$

the momentum conjugate to $\partial_\lambda \phi$ is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\lambda \phi)} = -\partial^\lambda \phi.$$

Insert the Lorentz variations:

$$\begin{aligned} J^\lambda &= (-\partial^\lambda \phi)\left(-\omega^\rho{}_\sigma x^\sigma \partial_\rho \phi\right) + \mathcal{L}\,\omega^\lambda{}_\sigma x^\sigma \\ &= \omega^\rho{}_\sigma x^\sigma\, \partial^\lambda \phi\, \partial_\rho \phi + \omega^\lambda{}_\sigma x^\sigma\, \mathcal{L}. \end{aligned}$$

3. Isolating the current for each generator. Because $\omega_{\mu\nu}$ is antisymmetric, it is convenient to rewrite $\omega^\rho{}_\sigma$ in terms of $\omega_{\mu\nu}$ and then factor out the independent parameters. We use

$$\omega^\rho{}_\sigma = \eta^{\rho\alpha}\omega_{\alpha\sigma},$$

and write the current as a linear combination of $\omega_{\mu\nu}$:

$$J^\lambda = \frac{1}{2}\,\omega_{\mu\nu}\, M^{\lambda\mu\nu},$$

for some tensor $M^{\lambda\mu\nu}$ antisymmetric in (μ,ν). To find $M^{\lambda\mu\nu}$, note that a general antisymmetric $\omega_{\mu\nu}$ can be written as a sum over basis matrices that generate boosts and rotations. Comparing

$$J^\lambda = \omega^\rho{}_\sigma \left(x^\sigma \partial^\lambda \phi\, \partial_\rho \phi\right) + \omega^\lambda{}_\sigma x^\sigma \mathcal{L}$$

with the general form $\frac{1}{2}\omega_{\mu\nu}M^{\lambda\mu\nu}$, we can guess the structure: it must be linear in x times the energy–momentum tensor $T^\lambda{}_\rho$. Recall the canonical energy–momentum tensor obtained previously,

$$T^\lambda{}_\rho = \partial^\lambda \phi\, \partial_\rho \phi - \delta^\lambda{}_\rho\, \mathcal{L}.$$

Consider the combination

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

This tensor is antisymmetric in (μ,ν) by construction:

$$M^{\lambda\nu\mu} = -M^{\lambda\mu\nu}.$$

Now form the current with $\omega_{\mu\nu}$:

$$\begin{aligned} J^\lambda &= \frac{1}{2}\,\omega_{\mu\nu}\, M^{\lambda\mu\nu} \\ &= \frac{1}{2}\,\omega_{\mu\nu}\left(x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}\right). \end{aligned}$$

Using the antisymmetry $\omega_{\mu\nu} = -\omega_{\nu\mu}$ and relabeling dummy indices, one can check that this reproduces the previous expression for J^λ. Thus we identify

$$\boxed{M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.}$$

4. Interpretation and conservation. Noethers theorem gives the conservation law

$$\partial_\lambda M^{\lambda\mu\nu} = 0,$$

which expresses conservation of total (orbital + spin) angular momentum associated with Lorentz invariance. In general, for fields with spin, there can be an extra "spin current" term $S^{\lambda\mu\nu}$ so that

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu} + S^{\lambda\mu\nu}.$$

For a *real scalar field*, there is no intrinsic spin, so $S^{\lambda\mu\nu} = 0$, and the angular-momentum current is purely *orbital*, given exactly by

$$M^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu}.$$

The corresponding conserved charges are

$$J^{\mu\nu} = \int d^3x\, M^{0\mu\nu}(t, \vec{x}),$$

which generate Lorentz transformations (rotations and boosts) on the fields. In particular, the spatial components J^{ij} are the usual angular momentum operators, while J^{0i} generate boosts.

Problem 4. From Discrete Systems to Field Theory (15 points)

Starting from the Lagrangian of a one-dimensional chain of coupled oscillators

$$L = \sum_n \left[\frac{1}{2}\mu\dot{\eta}_n^2 - \frac{\lambda}{2}(\eta_{n+1} - \eta_n)^2 - \frac{1}{2}\sigma\eta_n^2 \right],$$

derive its continuum limit as the lattice spacing $a \to 0$, obtaining a field theory in terms of a continuous field $\eta(x,t)$. Identify the effective mass m and propagation speed v, and discuss under what condition this theory becomes relativistic.

Start from

$$L = \sum_n \left[\frac{1}{2}\mu\dot{\eta}_n^2 - \frac{\lambda}{2}(\eta_{n+1} - \eta_n)^2 - \frac{1}{2}\sigma\eta_n^2 \right].$$

Let $x = na$ and $\eta_n(t) \to \eta(x,t)$. For small lattice spacing a,

$$\eta_{n+1} - \eta_n \approx a\,\partial_x \eta,$$

and the sum becomes an integral

$$\sum_n \to \frac{1}{a}\int dx.$$

Then

$$L = \int dx \left[\tfrac{1}{2}\mu(\partial_t\eta)^2 - \tfrac{1}{2}\lambda a^2(\partial_x\eta)^2 - \tfrac{1}{2}\sigma\eta^2 \right].$$

Define the continuum field theory Lagrangian density

$$\mathcal{L} = \tfrac{1}{2}\mu(\partial_t\eta)^2 - \tfrac{1}{2}\lambda a^2(\partial_x\eta)^2 - \tfrac{1}{2}\sigma\eta^2.$$

We can identify

$$v^2 = \frac{\lambda a^2}{\mu}, \qquad m^2 = \frac{\sigma}{\mu}.$$

Hence, in the continuum limit,

$$\boxed{\mathcal{L} = \tfrac{1}{2}(\partial_t\eta)^2 - \tfrac{1}{2}v^2(\partial_x\eta)^2 - \tfrac{1}{2}m^2\eta^2.}$$

The equation of motion is

$$\partial_t^2\eta - v^2\partial_x^2\eta + m^2\eta = 0.$$

The theory becomes **relativistic** when $v = 1$, so that time and space derivatives enter symmetrically.